Faris Hamid Sirraj

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Education

Boston University (BU) — *Boston, MA*
Master of Science in Robotics & Autonomous Systems — *September 2025 – December 2026*
Relevant Coursework: Deep Learning, Embedded Systems

National University of Singapore (NUS) — *Singapore*
Bachelor of Engineering in Computer Engineering (Honours) — *August 2021 – April 2025*

Experience

Curium Pte. Ltd. — *Singapore*
Robotics Software Engineer — *May 2023 – April 2025*

- Automated multi-sensor calibration workflows for autonomous trucks using an edge-compute-powered solution to eliminate manual processes and reduce calibration time by **90%**.
- Designed and deployed a secure Over-The-Air (OTA) firmware update pipeline using the AWS Suite, significantly reducing deployment cycles from several days to **under 2 hours**.
- Led deployment of containerized edge compute systems across 3+ smart city sites in Asia and Europe, ensuring stable, real-time performance in production conditions.

Software Engineer Intern — *October 2022 – December 2022*

- Analyzed and extracted sensor data from *ROS* bag and *eCal* recordings, reducing diagnostic time by **60%**.
- Developed end-to-end data extraction workflows, completely eliminating manual processing.
- Implemented optimized serialization protocols, improving data extraction speed by **75%**.

Publications

Faris Hamid Sirraj, Lee Yan Hui, Kutluhan Buyukburc, Anish M. Rao, Ali Hasnain. *Semantic-Aware Automated and Targetless Calibration of LiDARs for Smart City Infrastructure.* SPIE Future Sensing Technologies 2025 — Abstract accepted (Full paper submission due October 2025)

Notable Industry Projects

Targetless LiDAR Calibration in Urban & Industrial Environments — *May 2024 - July 2025*

- Engineered a targetless LiDAR calibration method that improved efficiency and accuracy in urban and industrial settings, reducing calibration time to **under 3 minutes**.
- Used loss-based optimization methods to achieve less than **5%** rotational and translational errors.

System Resource Monitoring Framework for NVIDIA Jetson Orins — *January 2025 - July 2025*

- Deployed lightweight diagnostics on production NVIDIA Jetson Orins, integrating InfluxDB, Grafana dashboards, and Slack APIs to monitor CPU, GPU, memory, and power usage.
- Enabled real-time performance tracking and automated alerts for faster diagnostics and proactive issue resolution.

Over-The-Air (OTA) System for Autonomous Truck Firmware Updates — *January 2024 - May 2024*

- Designed and deployed a cloud-based OTA system to remotely deliver firmware updates to the calibration stack of autonomous trucks during maintenance.
- Built the system using AWS ECR, Greengrass, Code Build and Code Pipeline, ensuring automated CI/CD workflows and seamless integration with edge devices.

Edge Calibration Systems for Autonomous Trucks — *May 2023 - April 2024*

- Developed and deployed an edge calibration system using Docker & Kubernetes, for Ubuntu Core-based edge devices.
- Enhanced calibration accuracy for LiDARs, Cameras and Radars.

Honors & Awards

IEEE Control Systems Chapter Prize, Singapore — Awarded for LiDAR calibration research — *June 2025*

Skills

Programming: Python, C, C++,
Embedded / Edge Platforms: Linux, NVIDIA Jetsons
Monitoring & Tools: Git, Jira, Grafana, Slack APIs
Frameworks: ROS/ROS2, Docker, Kubernetes
Cloud: AWS (Code Build/Pipeline, Greengrass, ECR)
CAD: Fusion360, Blender